Diabetes Progression Analysis

Report Summary

This report explores the diabetes progression dataset and evaluates simple and multiple linear regression models. BMI, blood pressure and s5 show positive, statistically significant associations with progression in the fitted multiple model. The model explains about 48% of the observed variation in this sample; the results are associational, not causal, and should be interpreted with the diagnostic limitations discussed below.

Question 1. Understanding and Preparing the Data

What does one row represent?

Each row represents one patient in the diabetes dataset. The dataset contains 442 observations and 11 variables.

Which variable is the outcome?

The outcome variable is progression. It represents the quantitative measure of disease progression used in this analysis.

Which variables are predictors?

The predictor variables are age, sex, bmi, bp, s1, s2, s3, s4, s5, and s6. These variables will be examined to investigate their relationships with diabetes progression and their usefulness for predicting the outcome.

Which variables are numerical, and which are categorical in meaning?

The variables age, bmi, bp, s1, s2, s3, s4, s5, s6, and progression are numerical. The variable sex is numerically coded but represents categories, so it is treated as a categorical variable in the analysis.

Are there missing values, duplicated rows or obviously problematic values?

The initial data checks found no missing values and no duplicated rows. Specifically, all variables had zero missing values, and the dataset contained zero duplicated rows. Further checks and visualisations will be used to examine the data and identify any other potential issues.

Why do health-related data require careful and responsible interpretation?

Health-related data require careful interpretation because statistical associations do not necessarily imply causation. The results describe patterns in the available dataset

and should not be treated as proof that a particular characteristic causes diabetes progression. Conclusions should also acknowledge the limitations of the data and the predictive models.

Question 2. Explore the Data

What are the most important characteristics of the dataset before modelling begins?

The dataset contains 442 patients and includes measurements of diabetes progression, age, BMI, blood pressure and six serum measurements. Diabetes progression has a mean of 152.13 and a median of 140.5. BMI has a mean of 26.38 and a median of 25.7, while the mean age is 48.52 years. These values provide an initial overview of the patients and the variation in the measurements.

Are the distributions approximately symmetric or clearly skewed?

The age distribution is relatively concentrated around middle age, although it is not perfectly symmetric. The BMI distribution appears moderately right-skewed, with a smaller number of patients having higher BMI values. The diabetes progression distribution also shows some right-skewness, as the mean (152.13) is higher than the median (140.5).

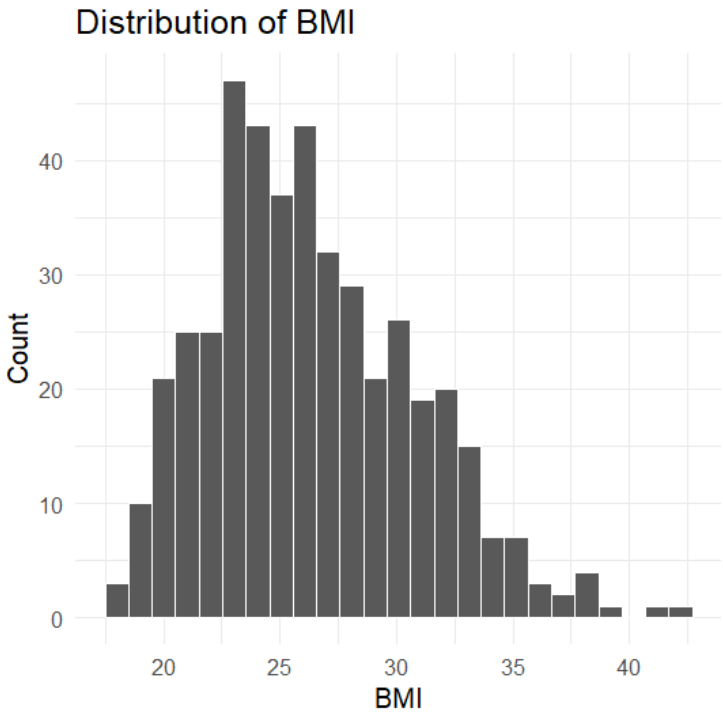


Figure 1. Distribution of BMI

Are there unusual observations?

The histograms show a small number of observations at relatively high values, particularly for BMI and diabetes progression. These observations may be unusual compared with the majority of the sample, but the histograms alone do not establish that they are errors or statistical outliers. Further investigation is needed before deciding how to handle them.

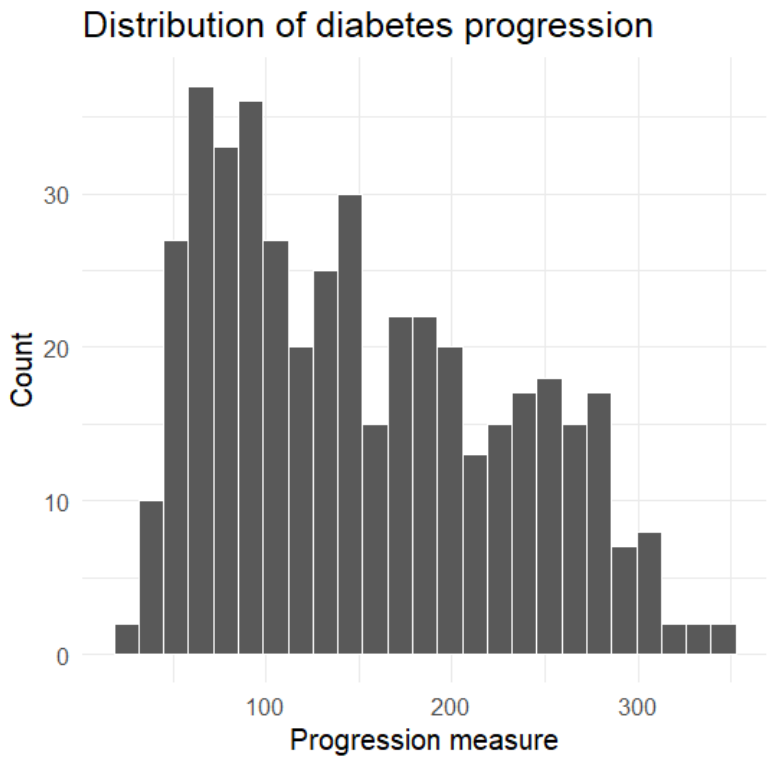


Figure 2. Distribution of Diabetes Progression

Which features of the data matter before modelling begins?

Before modelling, it is important to examine the distributions and variability of the outcome and predictor variables, identify potentially unusual observations, and investigate relationships between predictors. The sex variable should be treated as categorical rather than continuous, even though it is numerically coded. Possible correlations between predictors should also be checked because strong relationships between them may affect multiple regression models.

Question 3. Investigate Relationships

Which predictors are most strongly associated with progression?

BMI has the strongest positive correlation with diabetes progression (r = 0.5865), closely followed by s5 (r = 0.5659). Blood pressure (bp) and s4 also show moderate positive correlations (r = 0.4415 and r = 0.4305, respectively). In contrast, s3 has a negative correlation (r = -0.3948).

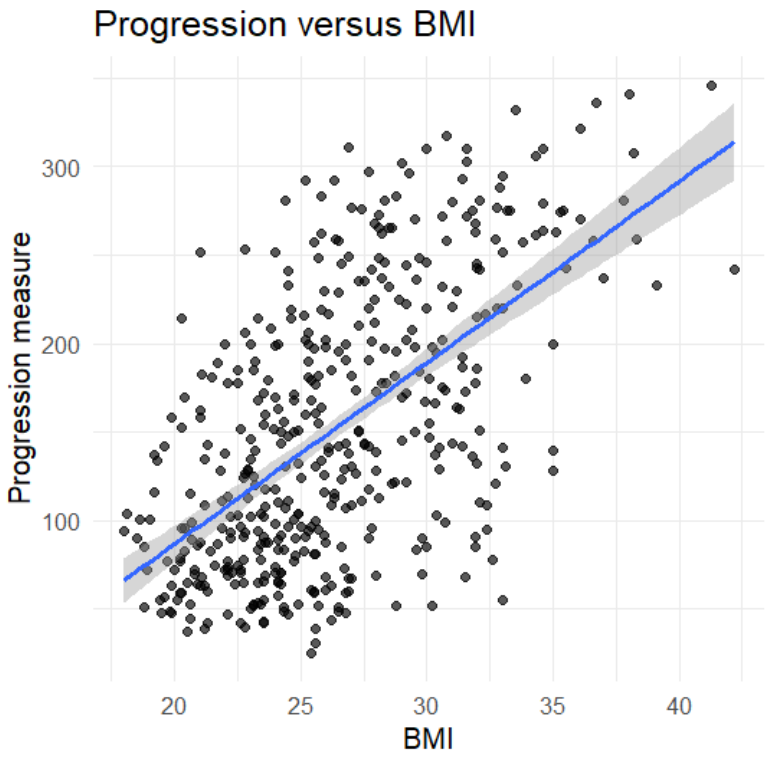


Figure 3. Diabetes Progression versus BMI

Which predictors are strongly related to each other?

The strongest relationship between predictors is between s1 and s2, with a correlation of r = 0.90. This indicates a very strong positive linear relationship between these two variables. Other notable correlations include s2 and s4 (r = 0.66), s4 and s5 (r = 0.62), and s1 and s4 (r = 0.54).

Could some predictors carry overlapping information?

Yes. The very strong correlation between s1 and s2 suggests that they may carry overlapping information. Other correlated pairs, such as s2 and s4, may also share some information. These relationships should be considered when fitting a multiple

regression model because strong correlations between predictors can cause multicollinearity.

Which variables look most promising for regression analysis, and why?

BMI and s5 appear most promising because they have the strongest positive correlations with diabetes progression. Blood pressure and s4 may also be useful because they show moderate positive correlations. However, their usefulness should be evaluated in the regression models, since correlations alone do not establish causation or show whether each predictor contributes independently.

Question 4. Build and Interpret Regression Models - Simple Linear Regression

The fitted regression equation is:

$$
\widehat{\text{progression}}=-117.773+10.233\cdot \text{BMI}
$$

The slope of 10.233 indicates that a one-unit increase in BMI is associated with an estimated increase of approximately 10.23 units in the diabetes progression measure, on average.

The intercept is −117.773, representing the predicted progression when BMI equals zero. This value has no useful practical interpretation in this context.

The model has an (R^2) of 0.344, meaning that BMI explains approximately 34.4% of the observed variation in diabetes progression. Therefore, BMI is informative for predicting progression, but a substantial proportion of the variation remains unexplained by this simple model.

Multiple Regression and Model Comparison

A multiple linear regression model was fitted using BMI, blood pressure (bp), and s5 as predictors of diabetes progression. The fitted equation was:

$$
\widehat{\text{progression}}=-334.881+6.500(\text{BMI})+0.903(\text{bp})+49.577(\text{s5})
$$

Holding blood pressure and s5 constant, a one-unit increase in BMI is associated with an estimated increase of 6.50 units in diabetes progression. Similarly, holding BMI and s5 constant, a one-unit increase in blood pressure is associated with an estimated increase of approximately 0.90 units. Holding BMI and blood pressure constant, a one-unit increase in s5 is associated with an estimated increase of approximately 49.58 units. All three coefficients are statistically significant at the 0.001 level.

The multiple model has an (R^2) of 0.480 and an adjusted (R^2) of 0.477. It therefore explains approximately 48.0% of the observed variation in diabetes progression, compared with 34.4% for the simple BMI-only model. The residual standard error also decreases from 62.5 to 55.8.

Overall, the multiple model provides a better fit to these data according to these measures. However, this does not establish causality or guarantee better predictive performance on new observations. The model assumptions and residual diagnostics should also be considered when interpreting the results.

Question 5. Evaluate the Analysis

Residuals versus fitted plot

The residuals are distributed both above and below zero. However, the smooth curve shows a slight curved pattern instead of remaining completely horizontal, suggesting that some non-linearity may be present. There are also several relatively large residuals. This indicates that the linear model may not capture all the structure in the data, so its assumptions should be interpreted with caution.

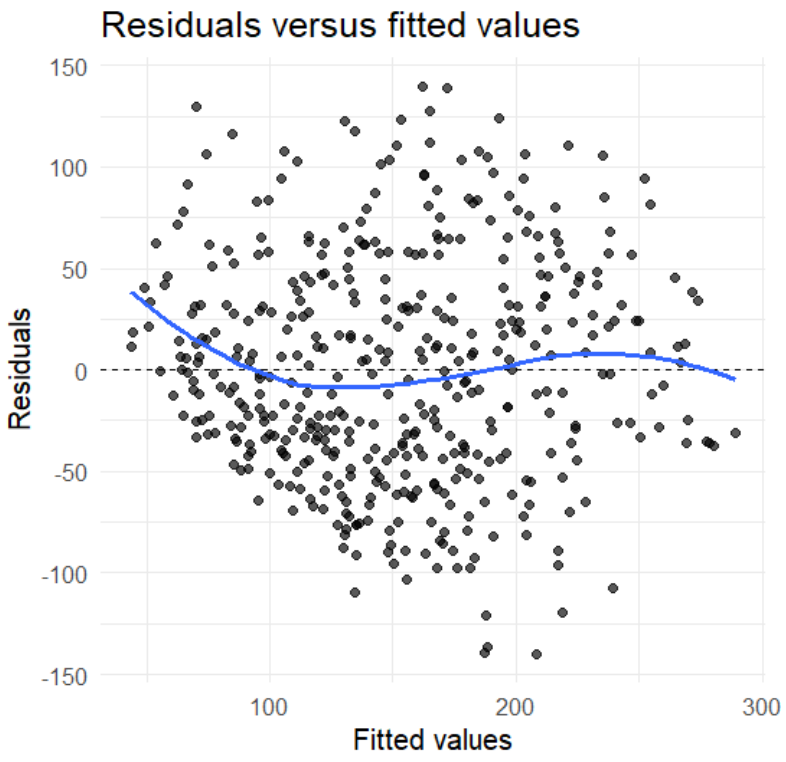


Figure 4. Residuals versus Fitted Values

Multicollinearity using VIF and the correlation matrix

The VIF values are 1.345 for BMI, 1.274 for blood pressure, and 1.342 for s5. Since all values are close to 1, there is little evidence of problematic multicollinearity among the predictors in the multiple regression model. The correlation matrix also shows that the selected predictors do not have extremely strong correlations with one another. Therefore, multicollinearity does not appear to be a major concern in this analysis.

Interpret at least two important coefficients

BMI: The estimated coefficient for BMI is 6.50, with a 95% confidence interval from 5.13 to 7.87 and a p-value of 5.55×10^{-19}. Holding blood pressure and s5 constant, a one-unit increase in BMI is associated with an estimated increase of 6.50 units in the diabetes progression measure. The confidence interval does not include zero, and the p-value is very small, indicating a statistically significant positive association.

Blood pressure: The estimated coefficient for blood pressure is 0.903, with a 95% confidence interval from 0.477 to 1.329 and a p-value of 3.74×10^{-5}. Holding BMI and s5 constant, a one-unit increase in blood pressure is associated with an estimated increase of 0.903 units in the progression measure. This coefficient is also statistically significant because its confidence interval excludes zero and its p-value is below 0.05.

Statistical significance versus practical importance

Statistical significance indicates that the observed associations are unlikely to be explained by random sampling variation alone under the model assumptions. However, statistical significance does not automatically imply practical importance. The practical importance of a coefficient depends on the scale of the predictor, the size of the expected change, and the context in which the result is interpreted. Therefore, although BMI and blood pressure are statistically significant predictors in this model, their practical importance should be assessed separately. These associations should not be interpreted as proof of causation.

Main statistical weaknesses and uncertainties

The analysis has several limitations. First, the residuals-versus-fitted plot shows a slight curved pattern, suggesting that the linearity assumption may not be fully satisfied. Second, some residuals are relatively large, indicating that the model does not predict every observation accurately. Third, the analysis relies on the selected predictors and the assumptions of linear regression. Other variables or alternative model specifications might explain additional variation in the outcome. Finally, statistical associations do not establish causation, and the results should be interpreted in the context of these limitations.

Question 6. Draw Appropriate Conclusions

Is the dataset experimental or observational?

The dataset is observational. The analysis examines existing measurements rather than assigning participants to experimental conditions or manipulating their characteristics.

Can the regression coefficients be interpreted causally?

No. The regression coefficients describe associations between the predictors and diabetes progression, holding the other included predictors constant. They do not establish that changes in BMI, blood pressure, or s5 cause changes in diabetes progression.

Could there be confounding variables not included in the dataset?

Yes. Other factors not included in the model may influence diabetes progression and may be associated with the predictors. Such omitted variables could affect the estimated regression coefficients.

Could measurement issues affect the results?

Yes. Measurement errors in the predictors or outcome could affect the estimated relationships. The analysis does not establish whether such errors are present, so this remains a potential limitation.

Would the results necessarily generalise to another population?

No. The findings may not generalise to populations with different characteristics or settings. Generalisation would require evidence that the dataset and study population are sufficiently representative of the target population.

Which conclusions are supported by the evidence?

In this dataset, BMI, blood pressure, and s5 have positive and statistically significant coefficients in the multiple regression model. The model has an R-squared of 0.480, meaning that it explains approximately 48% of the observed variation in diabetes progression in the analysed sample. The results support associations between these predictors and the outcome, conditional on the other predictors included in the model.

Which claims would go beyond the evidence?

It would go beyond the evidence to claim that BMI, blood pressure, or s5 directly cause changes in diabetes progression, that the model explains all relevant factors, or that its predictions will be equally accurate in every population. Statistical significance alone does not establish causality or practical importance.